

Youssef Chouay

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AI Engineer | Generative AI tools, LLM agents, Python systems, testing, documentation

EDUCATION

Master of Computer Science, Thesis

University of Ottawa

Expected Dec 2026

Ottawa, ON

- **Supervisor:** Vida Dujmovic | Awarded **\$52,000+** in research scholarships

Bachelor of Applied Science in Software Engineering

University of Ottawa

Dec 2024 | GPA: 9.0/10.0

Ottawa, ON

EXPERIENCE

Software Engineer Intern

Trend Micro / TrendAI

Jan 2026 – Present

Ottawa, ON

- Worked with the AI Security team to migrate a prompt-scan engine to state-of-the-art detection models for LLM threat analysis, including implementation, testing, debugging, and rollout support.
- Implemented chunking behavior for an inference engine in **Rust** and **Go**, improving support for long prompts and document-style inputs in AI security workflows.
- Owned design, implementation, testing, and AWS deployment of a backend service that retrieves company profiles and market data through **MCP**-based integrations for cybersecurity research workflows.

Artificial Intelligence Researcher

National Research Council Canada

May 2024 – Present

Ottawa, ON

- Worked with NRC, Delta Controls, and Carleton University to turn building-operations needs into AI-assisted BAS tools; published a **first-author peer-reviewed paper** and presented at **IEEE EPEC 2025**.
- Designed, tested, and deployed **Python/LangChain** agents connecting building automation streams with LLM tools, reducing estimated data-processing time and operator workload by **49%**.
- Built BAS ingestion and utility analytics pipelines for enterprise use with **SQLite** and time-series anomaly detection to flag usage spikes, baseload issues, and overnight leaks.

Software Engineer Intern

Wind River Systems

Sept 2022 – Aug 2023

Ottawa, ON

- Designed and implemented an automation dashboard using **Angular**, **TypeScript**, **Django**, and **PostgreSQL** to manage and analyze services used by customers like **NASA**, **Airbus**, and **Ford**.
- Improved dashboard query/API response times and UI responsiveness by **90%+** through optimizing API endpoints, database queries, and frontend rendering paths.

Software Developer

University of Ottawa

May 2022 – Apr 2024

Ottawa, ON

- Rebuilt and optimized the university search engine using **PHP**, **MySQL**, and **Apache**, improving response times by **80%** for **5,000+** students.
- Developed **PHP**, **Bash**, and Cron automation for indexing and migration workflows, improving search refresh speed by **54%**.

Teaching Assistant

University of Ottawa

Sept 2023 – Present

Ottawa, ON

- Supported graduate and undergraduate courses such as **Machine Learning for Bioinformatics**, Data Structures and Algorithms, programming paradigms, and discrete math through office hours, grading, tutorials, and student support.

PROJECTS

AtlasLLM | Founder / Personal Project

- Built and monetized a **multi-model AI chat platform** supporting **30+ LLMs** across 9 providers, generating nearly **\$10,000** in lifetime profit with **Next.js**, Convex, Redis, Cloudflare R2, and OpenRouter.
- Engineered a **custom DAG-based orchestration engine** for agentic research with auto-compaction support, context management, vectorized thread search, retries/timeouts, Langfuse tracing, and atomic **Redis Lua** credit deduction.

CLUI | macOS Developer Tool

- Built a macOS floating overlay for the **Claude Code CLI** using **Electron**, React, TypeScript, Zustand, and node-pty, with multi-tab sessions, global hotkeys, slash commands, attachments, and auto-updates.

NLP Phishing Detection | Bell Canada Research Project

- Built a phishing detection system using **NLP** and computer vision CNNs for website classification and clustering, achieving **98.4%** accuracy; shipped a Chrome extension and **AWS S3** retraining pipeline.

SKILLS

Languages: Python, TypeScript, Rust, Go, Java, C/C++, SQL, PHP, Bash, HTML, CSS

Generative AI: LLM agents, LangChain, MCP, prompt scanning, model orchestration, responsible AI, OpenRouter, Langfuse

AI/Data: PySpark, TensorFlow, NLP/CV, time-series anomaly detection, SQLite, PostgreSQL, Redis

Engineering: AWS, Next.js, Django, Electron, React, testing, debugging, technical documentation, Docker, Git